

Precision Pharmacology: Project Documentation & Notes

1. Project Overview & Business Value

The cost of bringing a single new drug to market is currently estimated at over \$2 billion, taking up to 10-15 years. However, "Drug Repurposing", finding new uses for existing or previously tested compounds, can drastically reduce this cost and timeline.

The goal of this project was to ingest millions of genetic data points, architect a relational SQL database, and identify "Precision Leads": chemical compounds that surgically alter disease genes without causing toxic off-target effects. Finally, the project evaluates the temporal behavior of these leads over time to classify them for specific clinical use cases (e.g., acute vs. chronic treatments) and automatically audits their Intellectual Property status.

2. Biological Context (Understanding the Data)

To understand the code, one must understand the biology it represents. This project utilizes the LINCS L1000 Dataset from the Broad Institute.

The Landmark Genes: Humans have roughly 20,000 genes. Measuring all of them is too expensive. The L1000 dataset only measures 978 "Landmark" genes. Because these genes are highly connected, if we know what these 978 are doing, we can infer the behavior of the rest of the cell.

MCF7 Cell Line: The experiments we filtered for were performed on MCF7 cells, which are a standard model for human breast cancer.

Z-Scores (The Metric of Impact): The values in our database are Z-scores. A Z-score of 0 means the drug did nothing. A Z-score of +3.0 means the drug strongly "turned on" (up-regulated) a gene. A Z-score of -3.0 means it strongly "turned off" (down-regulated) a gene. In biology, an absolute Z-score above 2.0 is considered statistically significant.

Temporal Dynamics (6h vs 24h): The dataset measures cellular responses at different time intervals. The 6-hour data captures the immediate, acute transcriptomic reaction to a drug, while the 24-hour data shows the mature, sustained downstream cascade as the cell adapts.

3. Data Engineering & Architecture

Overcoming the "Wide Data" Problem The raw data was provided in a .gctx format, resembling a massive spreadsheet with 978 rows (genes) and 14,861 columns (drug experiments). SQL databases perform terribly with 14,000 columns.

The Solution (Melt/Stack): Using Python's Pandas library, the data was "melted" into a "Long" format, generating over 14.5 million individual observations.

Dimensionality Reduction: We filtered out all Z-scores between -2.0 and 2.0, removing biological "noise" and reducing the dataset size to roughly 1 million significant interactions, ensuring high-speed SQL queries.

The Relational SQL Schema To emulate an enterprise data warehouse, the flat data was split into a Relational Schema using Primary and Foreign Keys:

`compound_metadata` (Table 1): Contains drug names, Mechanisms of Action (MoA), and structural chemistry strings (SMILES).

`genetic_signatures` (Table 2): Contains the actual experimental Z-scores.

All queries to answer the business questions utilized SQL JOIN statements to link the biological identity of the drug to its genetic impact.

4. Analytical Methodology (The Core Business Questions)

We utilized SQL to translate strategic pharmaceutical objectives into data queries:

Signature Reversal: Finding drugs that "flip" a known disease state (e.g., turning on genes that a disease turns off).

MoA Discovery: Matching the genetic fingerprints of unknown lab codes to well-understood, FDA-approved drugs.

Safety Screening: Counting extreme Z-scores ($|Z| > 4.0$) to flag drugs with high transcriptomic volatility (side effects).

Master Regulators: Identifying genes (like ATP6V1D) that react most frequently to chemical perturbations.

Precision vs. Power: Finding the "Goldilocks Zone" by charting the maximum potency of a drug against its specificity (how few genes it affects).

5. Temporal Pharmacodynamics & Clinical Triaging (Phase 5)

After identifying our top 10 most potent overall leads, we placed them under a temporal microscope by merging in the `pert_time` (perturbation time) metadata. By tracking the shift in both potency and the number of affected genes between the 6-hour and 24-hour marks, we triaged the candidates into distinct therapeutic profiles:

Cascade Inducers (Systemic Reset): Drugs that hit a specific target early (low gene count at 6h) but trigger a massive downstream metabolic shift by 24h. Ideal for chronic treatments like oncology.

Rapid Responders (Acute Intervention): Drugs that cause an immediate, broad biological shock (high gene count at 6h) that quickly refines and narrows by 24h as the cell adapts. Ideal for immediate, targeted interventions like inflammation suppression.

Efficacy Validation: The temporal analysis also verified that every single Top 10 lead reached the absolute maximum assay potency limit of 10.0 by the 24-hour mark, proving the accuracy of the Phase 1-4 selection algorithm.

6. Web Scraping & Intellectual Property (IP) Profiling

A critical question in pharmacology is Commercial Viability. We successfully identified highly potent drug leads, but we needed to determine their "Research Maturity." Are these drugs heavily studied (safe for repurposing), or are they undiscovered "Lab Codes" (untapped, highly lucrative Intellectual Property)?

The BeautifulSoup Web-Scraper: We engineered a custom web scraper using Python's requests and BeautifulSoup libraries.

The Literature Footprint Audit: The script systematically looped through our Top 10 Final Leads and physically navigated to the US National Library of Medicine (PubMed). It parsed the static HTML of the search results to extract the exact number of published, peer-reviewed scientific papers for each compound. This allowed us to mathematically categorize our leads into "Established Therapies" versus "Novel Proprietary IP" (0 published papers).

7. Architectural Considerations: The "Parallel Tracks" for Zero Data Loss

The current pipeline employs a "Discovery Funnel" approach: identifying the globally strongest drugs first, then analyzing their time-series behavior (Phase 5). While highly effective for generating a clean, executive narrative, this approach carries a critical data science assumption: it assumes that globally potent drugs are also the most potent at any given, isolated time point.

To ensure absolute **zero data loss** (e.g., catching a "late-blooming" drug that is completely dormant at 6h but becomes hyper-potent at 24h), future commercial iterations of this pipeline should implement a **"Parallel Tracks" architecture**. By embedding the temporal variable directly into the core SQL schema during Phase 1, the pipeline would branch to find the Top 10 drugs *exclusively* for 6-hour interventions, and a separate Top 10 *exclusively* for 24-hour interventions, ensuring no temporal outliers are lost to global averaging.